

DBMS

Lab Assignment-1

1. Display the data in existing default employees or emp or employee table.
QUERY: `SELECT * FROM emp;`
2. Display the structure of the above table.
QUERY: `DESC emp;`
3. Create a table “**Student**” with following structure:

Roll	Name	Age	Course	Math	Physics	Computer	Birthday
------	------	-----	--------	------	---------	----------	----------

Details of Attributes:-

Roll	Number (5)
Name	Varchar2 (30)
Age	Number (5)
Course	Varchar2 (5)
Math	Number (6, 2)
Physics	Number (6, 2)
Computer	Number (6, 2)
Birthday	Date

QUERY: `CREATE TABLE Student( Roll INT(5), Name VARCHAR(30), Age INT(5), Course VARCHAR(5), Math DECIMAL(6,2), Physics DECIMAL(6,2), Computer DECIMAL(6,2), Birthday DATE);`

```
mysql> CREATE TABLE Student( Roll INT(5), Name VARCHAR(30),  
-> Age INT(5), Course VARCHAR(5), Math DECIMAL(6,2), Physics DECIMAL (6,2), Computer DECIMAL(6,2), Birthday DATE);  
Query OK, 0 rows affected, 2 warnings (0.115 sec)
```

4. Create table **MSc** from the **Student** table with the same fields and same structure but without any data.
QUERY: `CREATE TABLE MSc AS SELECT * FROM Student WHERE 1=2;`

```
mysql> CREATE TABLE MSc AS SELECT * FROM Student WHERE 1=2;  
Query OK, 0 rows affected (0.123 sec)  
Records: 0 Duplicates: 0 Warnings: 0
```

5. Display the structure of **MSc** table.
QUERY: `DESC MSc;`

```
mysql> DESC MSc;
```

Field	Type	Null	Key	Default	Extra
Roll	int	YES		NULL	
Name	varchar(30)	YES		NULL	
Age	int	YES		NULL	
Course	varchar(5)	YES		NULL	
Math	decimal(6,2)	YES		NULL	
Physics	decimal(6,2)	YES		NULL	
Computer	decimal(6,2)	YES		NULL	
Birthday	date	YES		NULL	

```
8 rows in set (0.036 sec)
```

6. Create table 'MCA' from the Student table with the same fields and same structure but without any data. Rename Course with Department and Name with First Name.

QUERY: `CREATE TABLE MCA AS SELECT * FROM Student WHERE 1=2;`

```
mysql> CREATE TABLE MCA AS SELECT * FROM Student WHERE 1=2;
Query OK, 0 rows affected (0.155 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

`ALTER TABLE MCA RENAME COLUMN Course to Department;`

```
mysql> ALTER TABLE MCA RENAME COLUMN Course TO Department;
Query OK, 0 rows affected (0.289 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

`ALTER TABLE MCA RENAME COLUMN Name to First_Name;`

```
mysql> ALTER TABLE MCA RENAME COLUMN Name to First_Name;
Query OK, 0 rows affected (0.077 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

7. Display the structure of MCA table.

QUERY: `DESC MCA;`

```
mysql> DESC MCA;
```

Field	Type	Null	Key	Default	Extra
Roll	int	YES		NULL	
First_Name	varchar(30)	YES		NULL	
Age	int	YES		NULL	
Department	varchar(5)	YES		NULL	
Math	decimal(6,2)	YES		NULL	
Physics	decimal(6,2)	YES		NULL	
Computer	decimal(6,2)	YES		NULL	
Birthday	date	YES		NULL	

```
8 rows in set (0.018 sec)
```

8. Insert following records into the **Student** table:-

- 1, Rahul, 19, BCA, 79.5, 67, 89, 15-jun-93
- 2, Kunal, 21, BCA, 68, 76, 59.5, 16-aug-91
- 3, Aditi, 20, MSc, 90, 73, 56, 20-sep-92
- 4, Sumit, 20, MCA, 57.5, 78, 81, 07-dec-91
- 5, Anirban, 22, MCA, 80, 68, 63, 15-sep-94
- 6, Kumkum, 21, BCA, 72, 54.5, 60, 08-feb-95
- 7, Suman, 21, BCA, 91.5, 32, 61, 10-mar-94
- 8, Rohit, 22, MSc, 85, 76, 92, 19-apr-92

QUERY: **INSERT INTO Student VALUES**
 (1, 'Rahul', 19, 'BCA', 79.5, 67, 89, '1993-06-15'),
 (2, 'Kunal', 21, 'BCA', 68, 76, 59.5, '1991-08-16'),
 (3, 'Aditi', 20, 'MSc', 90, 73, 56, '1992-09-20'),
 (4, 'Sumit', 20, 'MCA', 57.5, 78, 81, '1991-12-07'),
 (5, 'Anirban', 22, 'MCA', 80, 68, 63, '1994-09-15'),
 (6, 'Kumkum', 21, 'BCA', 72, 54.5, 60, '1995-02-08'),
 (7, 'Suman', 21, 'BCA', 91.5, 32, 61, '1994-03-10'),
 (8, 'Rohit', 22, 'MSc', 85, 76, 92, '1992-04-19');

```
mysql> INSERT INTO Student VALUES
-> (1, 'Rahul', 19, 'BCA', 79.5, 67, 89, '1993-06-15'),
-> (2, 'Kunal', 21, 'BCA', 68, 76, 59.5, '1991-08-16'),
-> (3, 'Aditi', 20, 'MSc', 90, 73, 56, '1992-09-20'),
-> (4, 'Sumit', 20, 'MCA', 57.5, 78, 81, '1991-12-07'),
-> (5, 'Anirban', 22, 'MCA', 80, 68, 63, '1994-09-15'),
-> (6, 'Kumkum', 21, 'BCA', 72, 54.5, 60, '1995-02-08'),
-> (7, 'Suman', 21, 'BCA', 91.5, 32, 61, '1994-03-10'),
-> (8, 'Rohit', 22, 'MSc', 85, 76, 92, '1992-04-19');
Query OK, 8 rows affected (0.044 sec)
Records: 8 Duplicates: 0 Warnings: 0
```

9. Display all the students' details from **Student** table.

QUERY: **SELECT * FROM Student;**

```
mysql> SELECT * FROM Student;
```

Roll	Name	Age	Course	Math	Physics	Computer	Birthday
1	Rahul	19	BCA	79.50	67.00	89.00	1993-06-15
2	Kunal	21	BCA	68.00	76.00	59.50	1991-08-16
3	Aditi	20	MSc	90.00	73.00	56.00	1992-09-20
4	Sumit	20	MCA	57.50	78.00	81.00	1991-12-07
5	Anirban	22	MCA	80.00	68.00	63.00	1994-09-15
6	Kumkum	21	BCA	72.00	54.50	60.00	1995-02-08
7	Suman	21	BCA	91.50	32.00	61.00	1994-03-10
8	Rohit	22	MSc	85.00	76.00	92.00	1992-04-19

8 rows in set (0.007 sec)

10. Find out the details of the students with roll no 5 from **Student** table.

QUERY: **SELECT * FROM Student WHERE Roll=5;**

```
mysql> SELECT * FROM Student WHERE Roll=5;
```

Roll	Name	Age	Course	Math	Physics	Computer	Birthday
5	Anirban	22	MCA	80.00	68.00	63.00	1994-09-15

1 row in set (0.010 sec)

11. Show the roll, name, marks of all subjects for all students from **Student** table.

QUERY: **SELECT Roll,Name,Math,Physics,Computer FROM Student;**

```
mysql> SELECT Roll,Name,Math,Physics,Computer FROM Student;
```

Roll	Name	Math	Physics	Computer
1	Rahul	79.50	67.00	89.00
2	Kunal	68.00	76.00	59.50
3	Aditi	90.00	73.00	56.00
4	Sumit	57.50	78.00	81.00
5	Anirban	80.00	68.00	63.00
6	Kumkum	72.00	54.50	60.00
7	Suman	91.50	32.00	61.00
8	Rohit	85.00	76.00	92.00

8 rows in set (0.008 sec)

12. Display all the student details belonging to the **BCA** department.

QUERY: **SELECT * FROM Student where Course = 'BCA';**

```
mysql> SELECT * FROM Student where Course = "BCA";
```

Roll	Name	Age	Course	Math	Physics	Computer	Birthday
1	Rahul	19	BCA	79.50	67.00	89.00	1993-06-15
2	Kunal	21	BCA	68.00	76.00	59.50	1991-08-16
6	Kumkum	21	BCA	72.00	54.50	60.00	1995-02-08
7	Suman	21	BCA	91.50	32.00	61.00	1994-03-10

```
4 rows in set (0.010 sec)
```

13. Insert data in the 'MCA' table from 'Student' table where course is MCA.

QUERY: `INSERT INTO MCA SELECT * FROM Student WHERE Course = 'MCA';`

```
mysql> INSERT INTO MCA
-> SELECT *
-> FROM Student
-> WHERE Course = 'MCA';
Query OK, 2 rows affected (0.045 sec)
Records: 2 Duplicates: 0 Warnings: 0
```

14. Insert data in the 'MSc' table from 'Student' table where course is MSc.

QUERY: `INSERT INTO MSc SELECT * FROM STUDENT WHERE Course = 'MSc';`

```
mysql> INSERT INTO MSc SELECT * FROM STUDENT WHERE Course = "MSc";
Query OK, 2 rows affected (0.064 sec)
Records: 2 Duplicates: 0 Warnings: 0
```

15. Display the structure of 'Student' and 'MCA' table.

QUERY: `DESC Student;`
`DESC MCA;`

```
mysql> DESC Student;
```

Field	Type	Null	Key	Default	Extra
Roll	int	YES		NULL	
Name	varchar(30)	YES		NULL	
Age	int	YES		NULL	
Course	varchar(5)	YES		NULL	
Math	decimal(6,2)	YES		NULL	
Physics	decimal(6,2)	YES		NULL	
Computer	decimal(6,2)	YES		NULL	
Birthday	date	YES		NULL	

8 rows in set (0.020 sec)

```
mysql> DESC MCA;
```

Field	Type	Null	Key	Default	Extra
Roll	int	YES		NULL	
First_Name	varchar(30)	YES		NULL	
Age	int	YES		NULL	
Department	varchar(5)	YES		NULL	
Math	decimal(6,2)	YES		NULL	
Physics	decimal(6,2)	YES		NULL	
Computer	decimal(6,2)	YES		NULL	
Birthday	date	YES		NULL	

8 rows in set (0.015 sec)

16. Display all the details of each student in 'Student' table with Course appearing first in the display column list.

QUERY: `SELECT Course,Roll,Name,Age, Math,Physics,Computer, Birthday FROM Student;`

```
mysql> SELECT Course,Roll,Name,Age, Math,Physics,Computer, Birthday FROM Student;
```

Course	Roll	Name	Age	Math	Physics	Computer	Birthday
BCA	1	Rahul	19	79.50	67.00	89.00	1993-06-15
BCA	2	Kunal	21	68.00	76.00	59.50	1991-08-16
MSc	3	Aditi	20	90.00	73.00	56.00	1992-09-20
MCA	4	Sumit	20	57.50	78.00	81.00	1991-12-07
MCA	5	Anirban	22	80.00	68.00	63.00	1994-09-15
BCA	6	Kumkum	21	72.00	54.50	60.00	1995-02-08
BCA	7	Suman	21	91.50	32.00	61.00	1994-03-10
MSc	8	Rohit	22	85.00	76.00	92.00	1992-04-19

8 rows in set (0.007 sec)

17. Update the Math marks of the student with Roll no 7 from 91 to 95 in the 'Student' table.

QUERY: `UPDATE Student SET Math = 95 WHERE Roll=7;`

```
mysql> UPDATE Student SET Math = 95 WHERE Roll=7;  
Query OK, 1 row affected (0.028 sec)  
Rows matched: 1  Changed: 1  Warnings: 0
```

18. Update the Name of the student with Roll no 4 from Sumit to Sumitava in 'MCA' table.

QUERY: `UPDATE MCA SET First_Name = "Sumitava" WHERE Roll=4;`

```
mysql> UPDATE MCA SET First_Name = "Sumitava" WHERE Roll=4;  
Query OK, 1 row affected (0.021 sec)  
Rows matched: 1  Changed: 1  Warnings: 0
```

19. Delete the details of the student with Roll no 2 from the 'Student' table.

QUERY: `DELETE FROM STUDENT WHERE Roll=2;`

```
mysql> DELETE FROM STUDENT WHERE Roll=2;  
Query OK, 1 row affected (0.017 sec)
```

20. Delete the details of all students from the 'Student' table.

QUERY: `DELETE FROM STUDENT;`

```
mysql> DELETE FROM STUDENT;  
Query OK, 7 rows affected (0.041 sec)
```